20. Friday, August 2nd. Compactness

Definition: Let X be a Top. space. Let $\mathcal{U} \subseteq \mathcal{P}(X)$ be a family of subsets of X.

·) We say that $\mathcal{U}$ is a cover if $X = \bigcup_{U \in \mathcal{U}} U$.

·) We say that $\mathcal{U}$ is an open cover if each $U \in \mathcal{U}$ is open

·) A subcover of $\mathcal{U}$ is a subfamily $\mathcal{V} \subseteq \mathcal{U}$ that still covers X (i.e $X = \bigcup_{V \in \mathcal{V}} V$)

Examples: 1) In $\mathbb{R}^n$, $\mathcal{U} = \{ B(0,k) \mid k \in \mathbb{N} \}$ is an open cover: $\mathbb{R}^n = \bigcup_{k \in \mathbb{N}} B(0,k)$

Have subcovers, e.g $\mathbb{R}^n = \bigcup_{k \text{ even}} B(0,k)$ and $\mathbb{R}^n = \bigcup_{k \text{ odd}} B(0,k)$.

2) In $\mathbb{R}$, $\mathcal{U} = \{ (n-1, n+1) \mid n \in \mathbb{Z} \}$ forms an open cover.

However, no subfamily of $\mathcal{U}$ gives a cover ($n \in (m-1, m+1) \iff n = m$).

Definition: A Top space X is compact if every open cover of X admits a finite subcover.

Remark: 1) This says that for any collection $\{ U_i \mid i \in \Lambda \}$ of open sets with $X = \bigcup_{i \in \Lambda} U_i$ there are some indices $i_1, \dots, i_n \in \Lambda$ such that $X = U_{i_1} \cup \dots \cup U_{i_n}$

2) If X and Y are homeomorphic. Then X is compact if and only if Y is compact

Lemma 1: A subspace $Y \subseteq X$ is compact if and only if for every family $\{U_i \mid i \in \Lambda\}$ of open sets in X with $Y \subseteq \bigcup_{i \in \Lambda} U_i$, there exist $i_1, \dots, i_n \in \Lambda$ such that $Y \subseteq U_{i_1} \cup \dots \cup U_{i_n}$.

Proof: HW!

Examples

1) Any space with only finitely many open sets is compact.

Thus 1) any set with the trivial topology is compact

2) any finite set with any topology is compact.

2) A discrete space X is compact if and only if X is finite.

3) $\mathbb{R}$ is not compact: $\mathbb{R} = \bigcup_{n \in \mathbb{Z}} (n-1, n+1)$ does not admit a finite subcover.

Hence the intervals (a,b), $(-\infty, a)$ and (a, ∞) are not compact.

4) The interval $(0,1] \subseteq \mathbb{R}$ is not compact. The open cover $\left\{\left(\frac{1}{n}, 1\right] \mid n \in \mathbb{N}\right\}$ does not admit a finite subcover.

Hence the homeomorphic intervals $(a,b]$, $[a,b)$, $(-\infty, a]$ and $[a, \infty)$ are not compact

5) Similarly, $\mathbb{R}^n$ is not compact because $\mathbb{R}^n = \bigcup_{K \in \mathbb{N}} B(0,K)$ does not admit a finite subcover

(If $\mathbb{R}^n = B(0,K_1) \cup \dots \cup B(0,K_\ell)$ Then for $K = \max\{K_1, \dots, K_n\}$ we have

we get $\mathbb{R}^n = B(0,K)$, but we know e.g. $(K+1, 0, \dots, 0) \notin B(0,K)$

6) Let $X = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\} \subseteq \mathbb{R}$. Then X is compact. In fact given an open cover $\mathcal{U}$ of X, we know That There is $U \in \mathcal{U}$ with $0 \in U$. Since $\frac{1}{n} \to 0$, in $\mathbb{R}$ and U is a nbh of 0, we know That $\exists\, N \in \mathbb{N}$ such That $\frac{1}{n} \in U$ $\forall n \geq N$. The first $N-1$ terms $\frac{1}{1}, \frac{1}{2}, \dots, \frac{1}{N-1}$ each live in some element of $\mathcal{U}$, so we can get a subcover of X using N elements of $\mathcal{U}$.

Proposition 2: If $f: X \to Y$ is continuous and X is compact, Then $f(X)$ is compact.
As a consequence, any quotient of X is compact.

Proof: HW!

Examples: S^n, $\mathbb{RP}^2$, $S^1 \times S^1$ are compact.

Note: We will see That any $[a,b] \subseteq \mathbb{R}$ is compact. Note however That $(a,b) \subseteq [a,b]$ is not compact.

Proposition 3: If X is compact and $Y \subseteq X$ is closed, Then Y is compact.

Proof: Let $\mathcal{U}$ be a collection of open sets in X such that $Y \subseteq \bigcup_{U \in \mathcal{U}} U$.

Using that $Y \subseteq \bigcup_{U \in \mathcal{U}} U$, we get $X = Y \cup (X \setminus Y) = \bigcup_{U \in \mathcal{U}} U \cup (X \setminus Y)$

Since Y is closed, we know that $X \setminus Y$ is open in X.

Thus $\mathcal{U} \cup \{X \setminus Y\}$ is an open cover of X, which is compact.

Therefore $\exists\ U_1, \ldots, U_n \in \mathcal{U}$ such that

$$X = U_1 \cup \cdots \cup U_n \cup X \setminus Y.$$

Hence $Y \subseteq U_1 \cup \cdots \cup U_n$, as desired $\square$

Proposition 4: If X is Hausdorff and $Y \subseteq X$ is a compact subspace, Then Y is closed in X